

To:

The Editor-in-Chief
Neurocomputing
Elsevier B.V.

Date: September 5, 2026

Subject: Submission of original research manuscript titled “*Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models*”

Dear Editor-in-Chief,

I am pleased to submit the above-titled original research paper for consideration for publication in *Neurocomputing*.

The manuscript develops a common output-space read abstraction for finite attention, external retrieval, and parameter-expert routing. Its main contributions are:

1. a measure-theoretic definition with explicit existence conditions and support for query-dependent expert outputs;
2. exact ranked top- k bounds, an error-budgeted adaptive rule, and an approximate signed-cancellation bound; and
3. controlled synthetic and CounterFact studies of external memory, ANN candidates, routing, cancellation, and GPT-2 steering.

The experiments are intentionally presented as mechanism studies. Trainable synthetic experiments use five seeds, while the manuscript explicitly identifies the absence of large-scale ANN evaluation, competitive end-to-end model-editing baselines, and certified unlearning as limitations. Code, benchmark definitions, result files, and plotting scripts are available at <https://github.com/mossaiby/RoAS>.

I confirm that:

- the manuscript is original and is not under consideration elsewhere;
- the author declares no competing financial or personal interests; and
- the use of generative AI during manuscript preparation is disclosed in the manuscript according to Elsevier policy.

The paper may be relevant to readers interested in neural memory, retrieval, sparse routing, and mathematical descriptions of attention. Thank you for considering the manuscript.

Sincerely,

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GitHub Repository: <https://github.com/mossaiby/RoAS>